

Interpreting the CLIP Embedding Space with PCA and k-Means

By Nandini, Megha, Naomi, Dhanashree, Kirsten



Introduction

CLIP embeddings capture rich semantic information, but the structure of their embedding space is not well understood.

The principal components (PCs) of CLIP's embedding space may correspond to meaningful semantic or stylistic directions that can be systematically examined. Our project analyzes the CLIP embedding space by:

1. Performing PCA on CLIP embedding space on 3 different datasets that vary in semantic diversity.
2. For each dataset, we perturb CLIP embeddings along a few principal components.
3. Reconstruction of images after perturbation using Stable Diffusion.
4. Characterizing the change in images based on perturbation.
5. Performing K-means clustering on CLIP embeddings and pixel features of the images.
6. We observe how images were clustered differently in each case.

Motivations

The questions our project seeks to answer are:

1. Can we interpret what each principal component of the CLIP embedding space represents?
2. Is interpretability of CLIP embeddings affected by the specificity of the dataset?
3. How do perturbations along principal components change the features in an image?
4. How are the results of K-means clustering different when performed on CLIP embeddings compared to when performed on the pixel features of images.

Through this project, we analyze how well CLIP captures and organizes high-level semantic relationships between images.

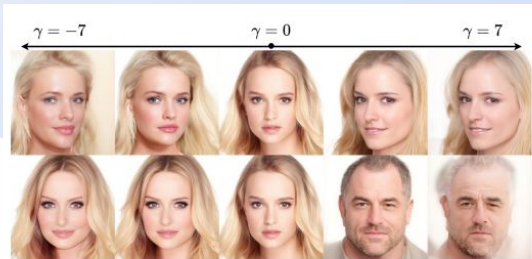
Related Work & Literature Review

Interpretability of Principal Components in semantic embedding space

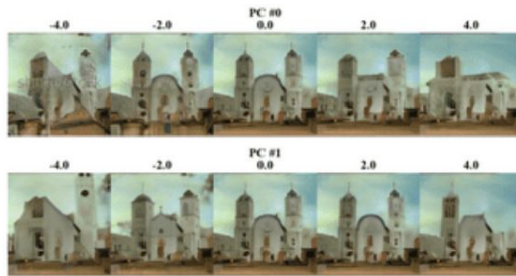
- Haas et al. (2024) used PCA on the semantic latent space of a diffusion model trained on face images to enable variation of identifiable features, including pose and gender.
- They found that less structured data, such as churches, produced “less interpretable principal directions.”

Semantic Image Clustering

- In a Journal of Electronics paper, CLIP was used to compute similarity scores between images of rooms and their text descriptions, from which they found similarity vectors that they put through K-Means clustering. This achieved clustering based on room characteristics and not just visual similarity (Yuda et al., 2025).



(a) Two dominant PCA directions



Haas et al. 2024

References

Haas, R., Huberman-Spiegelglas, I., Mulayoff, R., Graßhof, S., Brandt, S., & Michaeli, T. (2024). Discovering Interpretable Directions in the Semantic Latent Space of Diffusion Models. 18th International Conference on Automatic Face and Gesture Recognition. [aXiv/2303.11073v2](https://doi.org/10.2303/11073/v2)

Yuda, E., Morikawa, N., Kaneko, I., & Hirahara, D. (2025). CLIP-Guided Clustering with Archetype-Based Similarity and Hybrid Segmentation for Robust Indoor Scene Classification. Electronics, 14(23), 4571. <https://doi.org/10.3390/electronics14234571>

Datasets

Birds

Source: THINGS

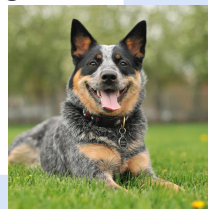
The Birds dataset contains about 300 images of a variety of birds. These images are a sample form the THINGS dataset. This dataset contains images of about 17 types of birds.



Animals

Source: THINGS

The Animals dataset contains about 1900 images of a variety of animals including birds. This dataset is sampled from the THINGS dataset. This dataset is more diverse in the category of images and is important for our analysis.



Patterns

Source : Patterns dataset on Kaggle

The Patterns dataset contains images of around 45 different textures and patterns. We randomly sampled 30 images from each of these categories to create our dataset.



Methods

We use three methods in our exploration of CLIP and its abilities. We use Principal Component Analysis and k-means to apply course concepts to our project. We also utilized the code provided by the teaching staff to implement a diffusion model to reconstruct images from CLIP embeddings after performing perturbations.

CLIP + Stable Diffusion

We pass in images through CLIP to retrieve embeddings that contain semantic information about the images.

We use a stable diffusion model to reconstruct the resulting images of our perturbations along principal components of the embeddings.

k-Means

We perform k-means clustering on CLIP embeddings and pixel vectors of the images in our datasets to observe how the images are clustered differently in each case.

PCA

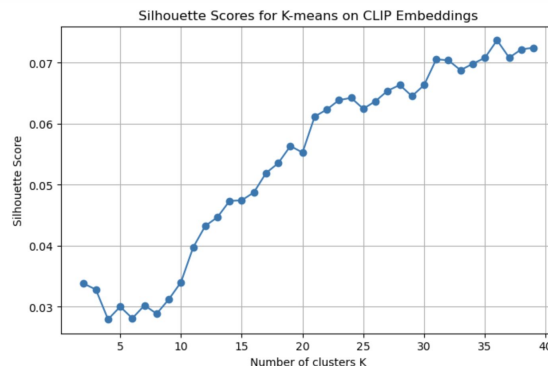
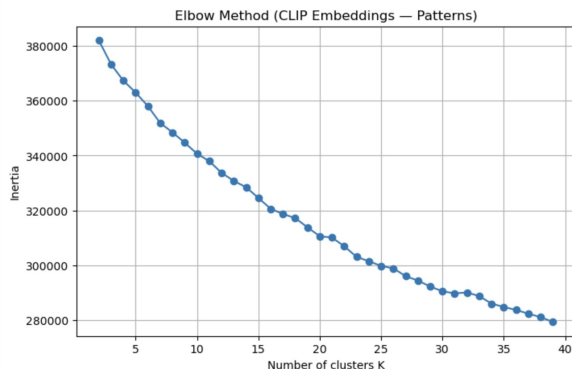
We apply Principal Component Analysis to the CLIP embeddings of our datasets.

We perform perturbations using the top three principal components to observe how the features of images vary along these directions.

k-Means

We performed **K-means clustering** on CLIP embeddings and pixel features of the images. We hypothesized that clusters formed from CLIP embeddings would group images by semantic similarity more effectively than clusters formed from pixel features.

In both cases, we initially used the **Elbow Method** and **Silhouette Score** to find a good estimate for the number of clusters. We ultimately decided to set the number of clusters to be the number of categories or subfolders in each dataset (e.g., we had 17 categories of birds, so we set $k=17$ for k-means on birds). By following this approach for both versions of clustering, we hoped to have more consistency in our implementation so that we can make more reasonable comparisons.



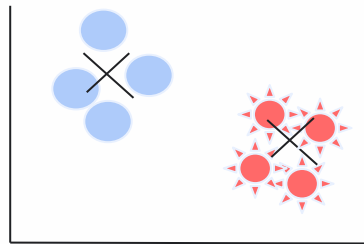
Note: We applied PCA to the pixel features of images to reduce dimensionality without compromising on image size. This was done to make the k-means clusters more stable.
[!\[\]\(8d0f0e0fe25b320c33272c52aec1fbca_img.jpg\) `sklearn-k-means-clustering`](#)

k-Means

For each dataset, we ran **K-means** on the CLIP embeddings and on the dimensionality-reduced pixel data. Then, we visually compared how groups were made with CLIP versus by pixel similarity.

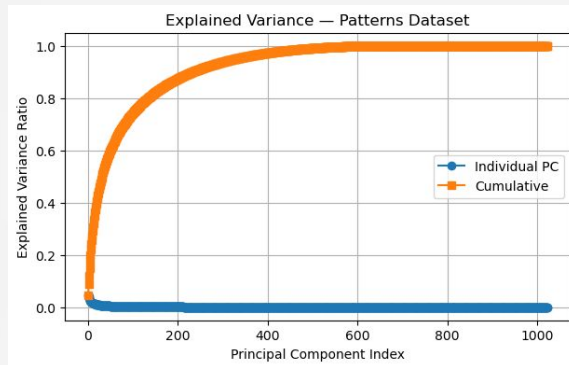
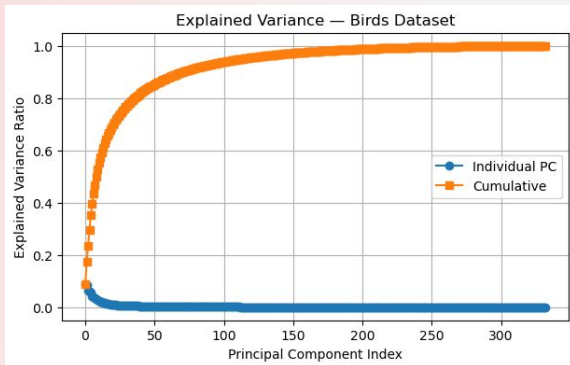
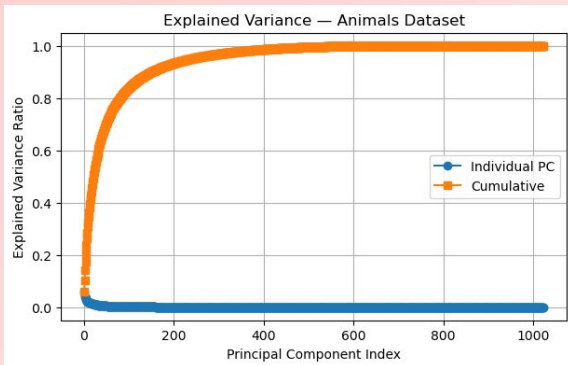
We also looked at the **cluster centers** (centroids) to see if, when creating images from them and visualizing them alongside their respective clusters, they are fairly representative of the unifying features of each grouping.

We used **StableDiffusion** to reconstruct images from the **mean CLIP embedding** of each cluster, and the **Pillow library** and array **reshaping** to turn the pixel centroids into images.

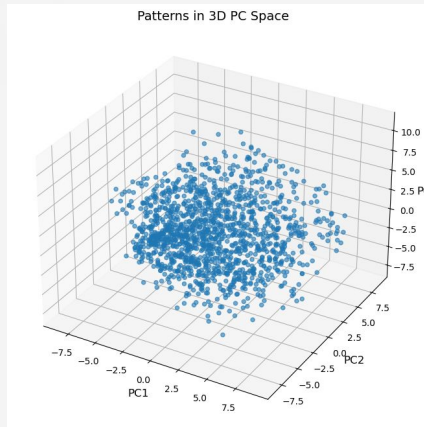
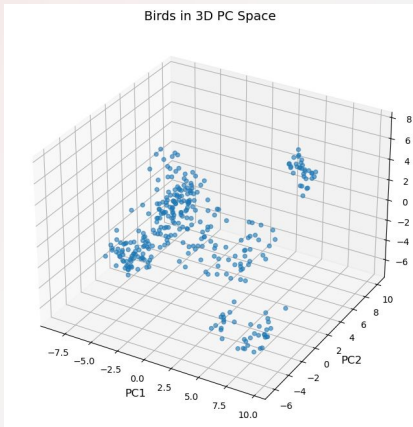
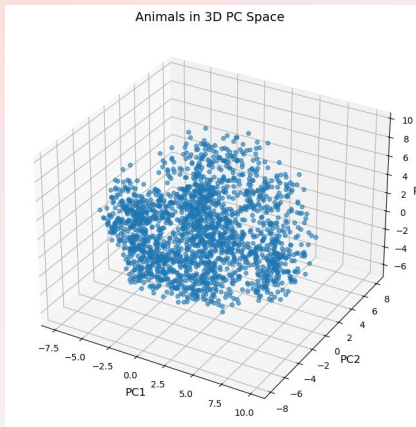


Performing PCA

Performed **PCA separately** on each of the 3 datasets, evaluated **scree plots**, and chose **top 3 PCs**.

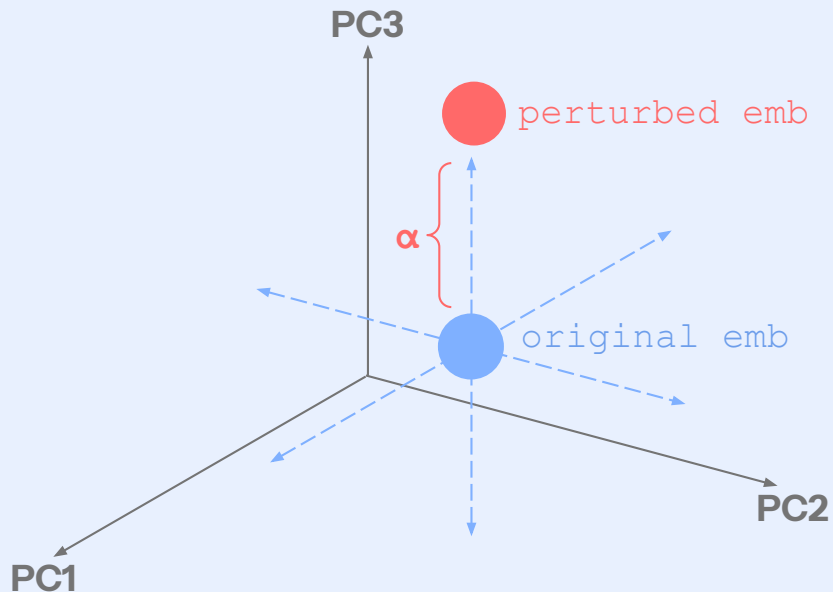


Scree plots



*Visualizing
embeddings in
3-PC space*

Controlled Perturbation



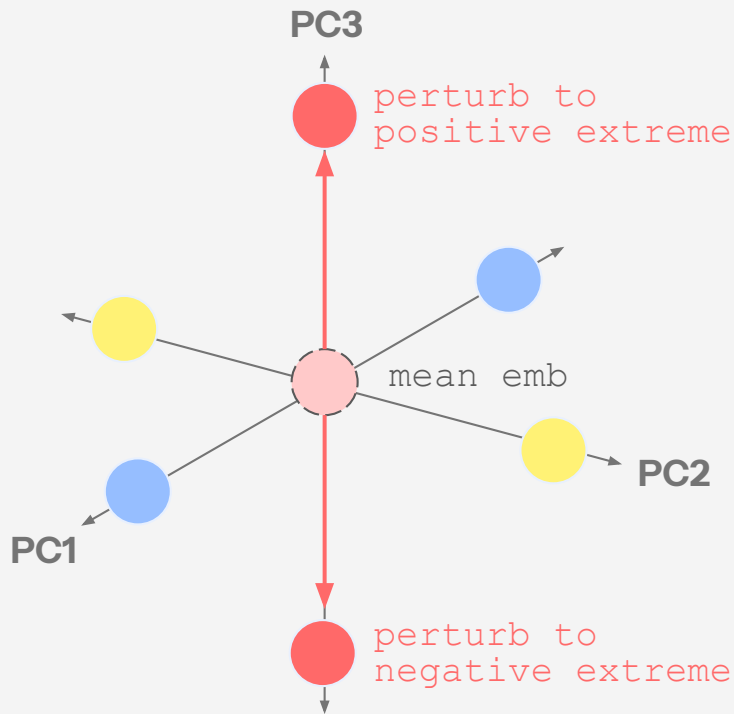
- Take CLIP embedding of **single image** from the dataset
- Slide embedding along each of the **3 PC axes** in small increments (α)
 - small α jumps = $[-3, -2.5, \dots, 0, \dots, 2.5, 3]$
 - medium α jumps = $[-20, -10, \dots, 0, \dots, 10, 20]$
 - large α jumps = $[-50, -20, \dots, 0, \dots, 20, 50]$
- **Reconstruct** perturbed embeddings with **StableDiffusion**
- Characterize changes from perturbations
- Characterize what **PC axes control**

$$\text{perturbed embedding} = \text{original embedding} + \alpha(\text{PC direction})$$

Synthetic Generation

To visualize **protoypical changes** along each PC axis, we perform perturbation on the **mean embedding** of the dataset.

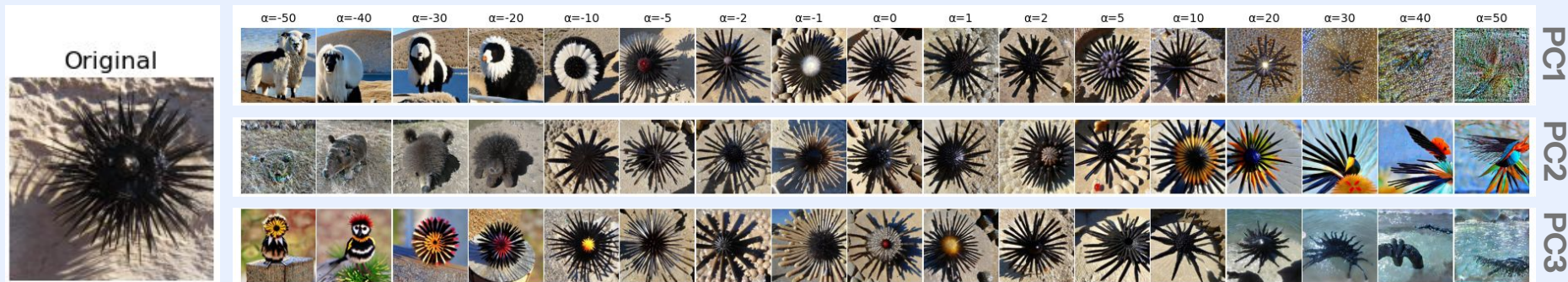
- Mean embeddings may be nonsensical but represent all data
- Perturbing shows us **typical feature changes** handled by each PC axis
- Extreme ends of each axis describe **archetypal images**
- Confirms results of individual-image experiments



Results



Results: Animals — Controlled Perturbation



- Original sea urchin transitions smoothly to large elk-like animal, to colorful bird-like
- Background changes slightly to match the animal it becomes: adding water, grass, etc.
- This pattern is observed for all images and a smooth transformation matrix forms.

Results: **Animals** — Synthetic Generation

PC1

 $\alpha=-20$  $\alpha=-10$  $\alpha=-5$  $\alpha=-3$  $\alpha=0$  $\alpha=3$  $\alpha=5$  $\alpha=10$  $\alpha=20$ 

Large, furry, land-dwelling

Small, hairless, insect-like

PC2

 $\alpha=-20$  $\alpha=-10$  $\alpha=-5$  $\alpha=-3$  $\alpha=0$  $\alpha=3$  $\alpha=5$  $\alpha=10$  $\alpha=20$ 

Medium-sized, boar-like, grassy habitat

Colorful, small, bird-like

PC3

 $\alpha=-20$  $\alpha=-10$  $\alpha=-5$  $\alpha=-3$  $\alpha=0$  $\alpha=3$  $\alpha=5$  $\alpha=10$  $\alpha=20$ 

Brown, small, bird-like

Smooth, aquatic, fish-like

mean

Results: Animals — k-Means Clustering

- **CLIP embeddings** clustered **same-species animals together**
 - Clusters contain just 1 or 2 same or highly similar species
- Latents of **raw pixel vectors** clustered **similar images together**
 - Clusters **less homogenous** in species
 - Grouped images more based on overall **colors, lighting, habitat**

k=123

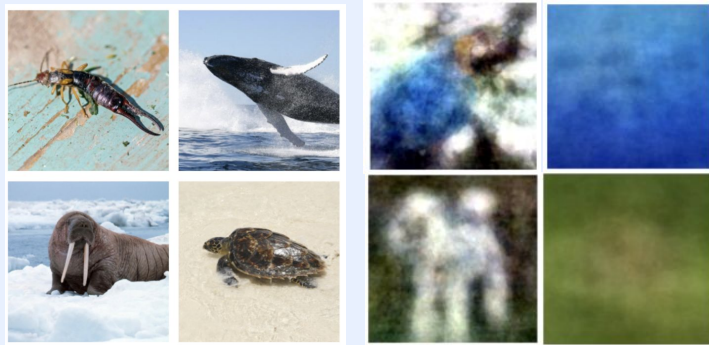
CLIP Clusters



Cluster members

Centroid recreation

Raw Pixel Clusters

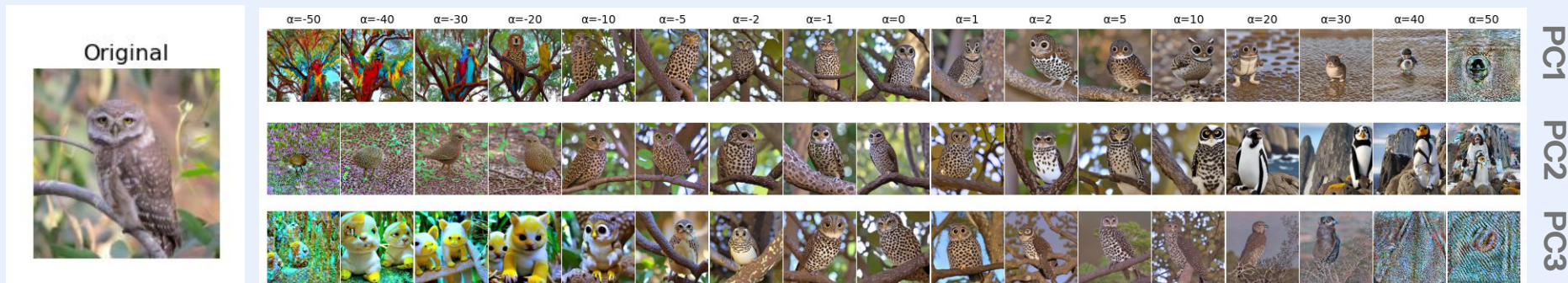


Cluster members

Centroid recreations

Takeaway: **CLIP embeddings** encode **semantic content of images** whereas **raw pixel vectors** capture **superficial visual characteristics**.

Results: **Birds** — Controlled Perturbation

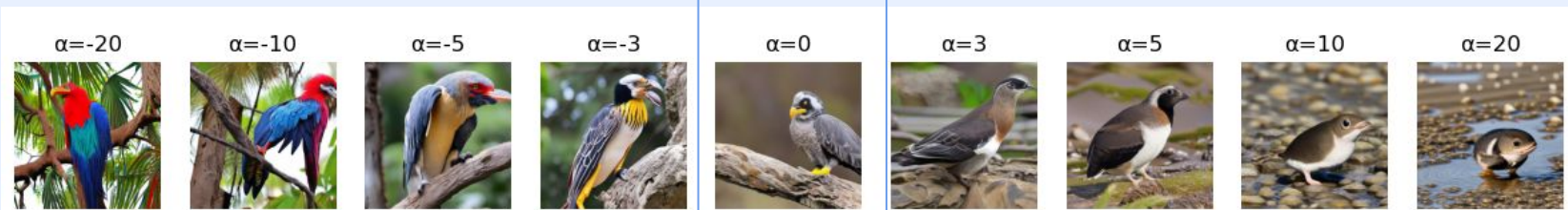


- Original owl transitions smoothly to different species of birds- parrots, ducks, penguins, etc
- Background changes slightly to match the bird it becomes: adding water, branches, etc.
- In PC3 the extremes really start to deviate from the original owl, splitting off into two creatures rather than one and becoming animals rather than staying a bird.
- Images near the ends in all PCs also start to turn into objects/abstract

More examples given in notebooks.

Results: Birds — Synthetic Generation

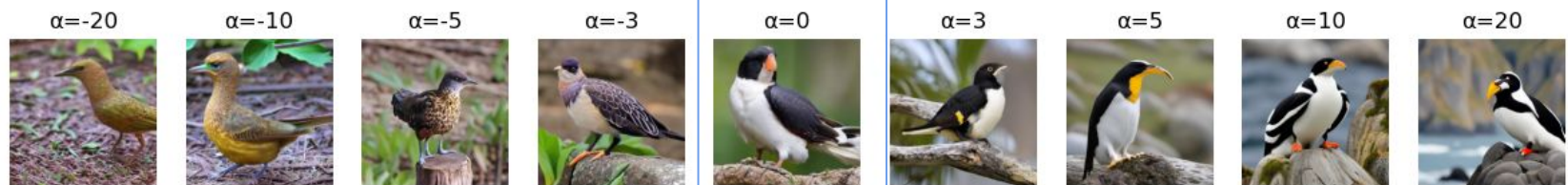
PC1



Larger, more colorful bird

Smaller bird with less diversity in color and background

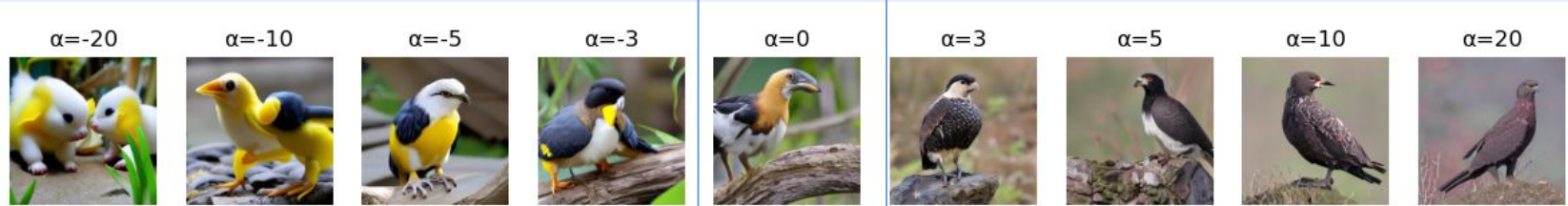
PC2



More colorful bird, different background

Similar bird but different background

PC3



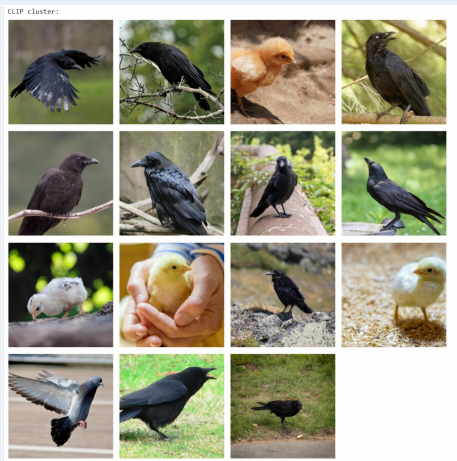
Two creatures, animal like

Singular, less variation in color

mean

Results: Birds — k-Means Clustering

CLIP Clusters



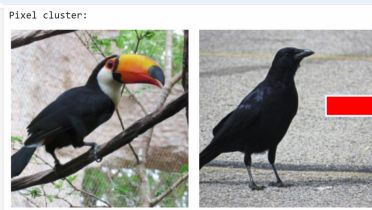
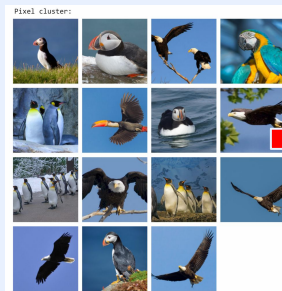
Cluster members

C5



Centroid recreation

Raw Pixel Clusters



Cluster members

k=17

C11



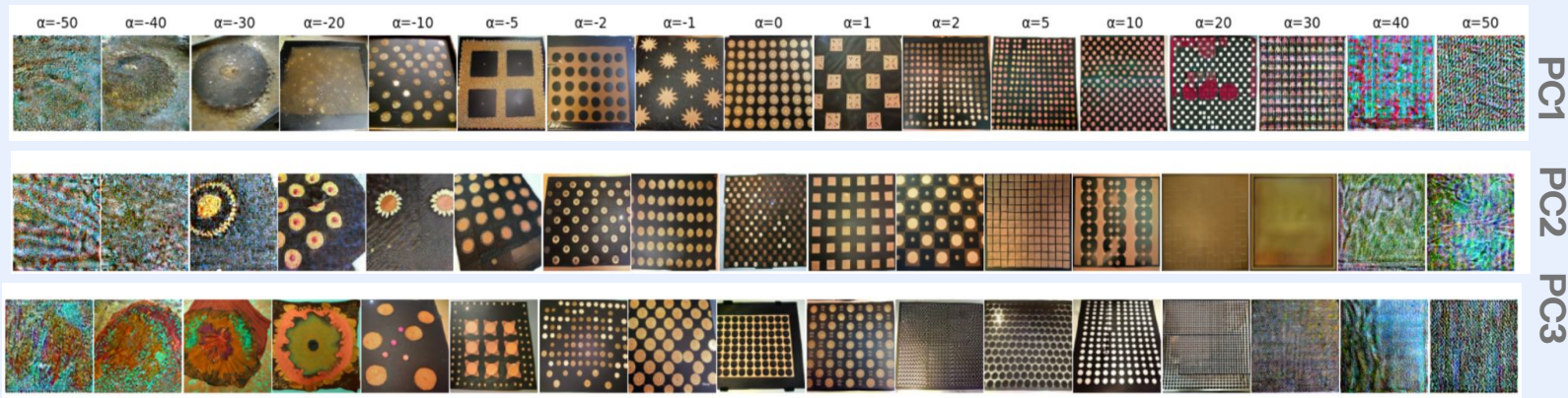
Centroid recreations

C2



Takeaway: Centroid clarity in pixel-space k-means directly depends on cluster size, because averaging only produces recognizable images when a small number of highly similar images are blended.

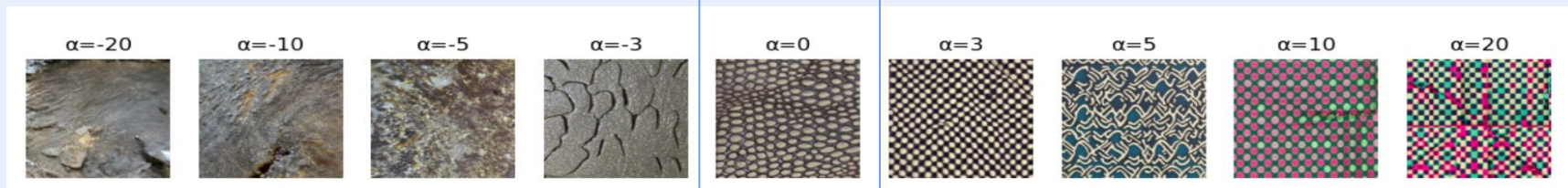
Results: Patterns— Controlled Perturbation



- Pattern maintains overall similarity in medium and color
- Differs in number, shape, and orientation of yellow figures

Results: Patterns— Synthetic Generation

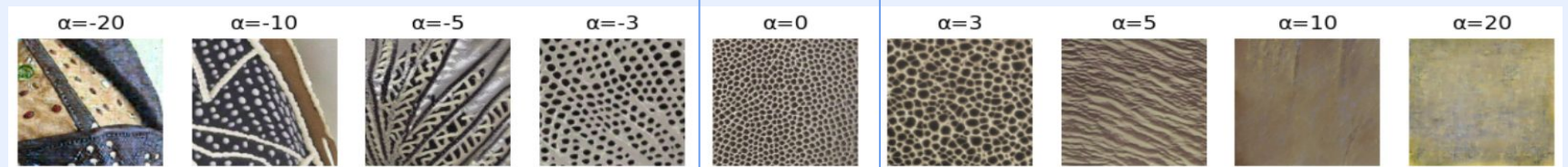
PC1



Less color, less detail

Colorful, fine patterns with clear shape

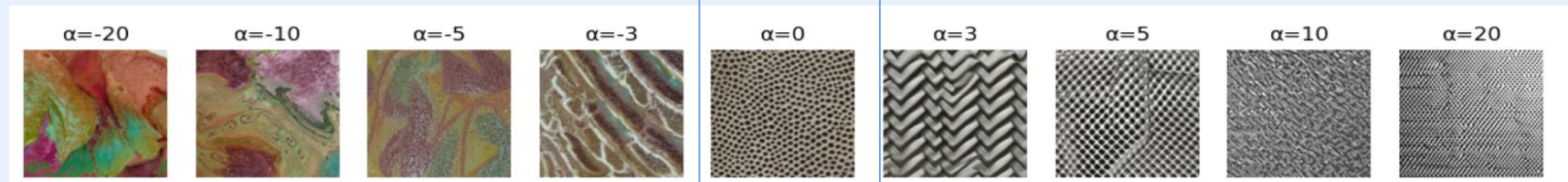
PC2



Different textures and new details

Brown, pattern fades

PC3



Colorful and abstract

Gray, texture becomes fine

mean

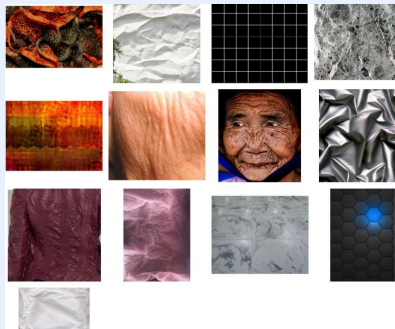
Results: Patterns— k-Means Clustering

- CLIP embeddings clustered similar **patterns** across different contexts
 - Clusters contain predominantly one pattern
- Latents of raw pixel vectors clustered images based on **color, shape, contrast**
 - Patterns were usually not consistent within clusters
 - Pattern representation in centroid images depended on number of images per cluster

k=45

CLIP Clusters

Cluster members

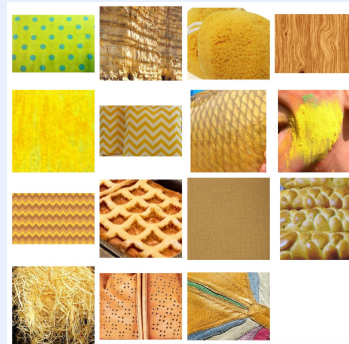


Centroid recreation

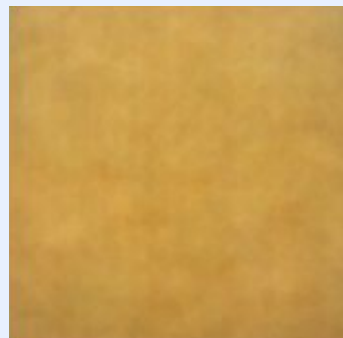


Raw Pixel Clusters

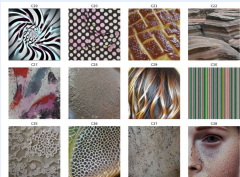
Cluster members



Centroid recreation



More centroid images



More centroid images



Discussion

As we've been answering our motivation questions along the way in our results, we'd like to add on some additional things we've learned:

We now know that Stable Diffusion is not deterministic, so the randomness in generated images made some of our clusters harder to interpret. This made it difficult to tell whether certain patterns came from real structure in the data or from noise. We also saw that the amount of variance in the dataset strongly affects how interpretable the axes are, and it would be useful to explore whether there is an ideal amount of variance that produces clearer results.

We also spent a very large amount of time exploring different methods as our main motivation was open exploration of CLIP space to strengthen our understanding of it. *The Elbow Method and Silhouette Scoring are some examples- even though the code for these experiments are not shown in the final presentation due to time constraints, you can find it in our notebooks.*

Further directions and improvements:

- Test cluster stability across multiple random seeds to see how much of the structure is reliable rather than noise.

Discussion (cont'd)

More extensions to our project could include:

- Along the PCs we examined, it wasn't always easy to categorize the feature changes we observed, and we saw multiple features changing. Research from **Haas et al.** suggests that there are ways to **disentangle features** in order to keep the rest of features the same while visualizing how one feature varies along a principal component. One extension to our project could be to **use models from their research** to isolate changes to specific features to explore how they change.
- Our methods did not generalize to all types of image data, and did not work as well for the patterns dataset since it behaved differently due to its lack of a single subject in each image and being made up of textures and fine details. We could improve our methods to **apply models that extract pattern features to be able to more accurately measure feature variations in patterns data.**
- Another further direction we were thinking about is analyzing using **ICA** and **exploring any differences in results.**